Georgios Alexopoulos

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Work Experience

National Infrastructures for Research and Technology (GRNET) 05/2023 - now

Systems Architect

Arrikto

Software Engineer 07/2021 - 04/2023

- Designed and implemented Kiwi, a GPU sharing/virtualization mechanism for Kubernetes and on-prem based on my diploma thesis
- Automated the build process of container images with an emphasis on caching and reproducibility

Software Engineer Intern 01/2020 - 06/2021

Education

2014 - 2021 Diploma (5-year degree - 300 ECTS - MEng equivalent) [GPA: 8.72/10]
School of Electrical and Computer Engineering, National Technical University of Athens

2023 - now Ph.D.
Department of Business Administration, National and Kapodistrian University of Athens
Subject: "Cross-Language Analysis of Software Ecosystems"
Supervisor: Dimitris Mitropoulos

Awards

2026 Outstanding Doctoral Research Award
Department of Business Administration, National and Kapodistrian University of Athens

Publications

1. Georgios Alexopoulos, Nikolaos Alexopoulos, Diomidis Spinellis, and Dimitris Mitropoulos. *When AI Agrees Too Much: Manipulating Automated Code Review via Bias Injection.* In Proceedings of the 49th International Conference on Software Engineering (ICSE '27), 2027. [PDF]

2. Georgios Alexopoulos, Thodoris Sotiropoulos, Zhendong Su, and Dimitris Mitropoulos. *Best of both worlds: Effective foreign bridge identification in V8 embedders for security analysis.* In Proceedings of the 47th IEEE Symposium on Security and Privacy (S&P '26), 2026. [PDF] [Code]

3. Georgios Alexopoulos, Thodoris Sotiropoulos, Georgios Gousios, Zhendong Su, and Dimitris Mitropoulos. *PyXray: Practical Cross-Language Call Graph Construction through Object Layout Analysis.* In Proceedings of the 48th International Conference on Software Engineering (ICSE '26), 2026. [PDF] [Code]

4. Georgios-Petros Drosos, Georgios Alexopoulos, Thodoris Sotiropoulos, Dimitris Mitropoulos, and Zhendong Su. *Unfulfilled Promises: LLM-Based Detection of OS Compatibility Issues in Infrastructure as Code.* In Proceedings of the ACM on Software Engineering (FSE '26), 2026. [PDF]

5. Georgios Alexopoulos, Nikolaos Alexopoulos, Thodoris Sotiropoulos, Charalambos Mitropoulos, Zhendong Su, and Dimitris Mitropoulos. *Cross-Ecosystem Vulnerability Analysis for Python Applications.* arXiv preprint arXiv:2603.18693, 2026. [arXiv]

6. Georgios Alexopoulos, Konstantinos Karakatsanis, Nikolaos Alexopoulos, Dimitris Mitropoulos, and Thodoris Sotiropoulos. *Reducing Cold-Start Latency in Serverless Applications via Dynamic Slicing.* arXiv preprint arXiv:2609.14040, 2026. [arXiv]

7. Konstantinos Karakatsanis, Georgios Alexopoulos, Ioannis Karyotakis, Foivos Timotheos Proestakis, Evangelos Talos, Panos Louridas, and Dimitris Mitropoulos. *PyTrim: A practical tool for reducing Python dependency bloat.* In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE '25): Demonstrations Track, 2025. [PDF] [Code]

8. Georgios-Petros Drosos, Thodoris Sotiropoulos, Georgios Alexopoulos, Dimitris Mitropoulos, and Zhendong Su. *When Your Infrastructure is a Buggy Program: Understanding Faults in Infrastructure as Code Ecosystems.* In Proceedings of the ACM on Programming Languages (OOPSLA '24), 2024. [PDF] [Code]

9. Georgios Alexopoulos and Dimitris Mitropoulos. *nvshare: Practical GPU sharing without memory size constraints.* In Proceedings of the 46th International Conference on Software Engineering (ICSE '24): Demonstrations Track, 2024. [PDF] [Code]

Software

nvshare Practical GPU sharing for Kubernetes and Linux. Lets multiple processes or containers run on a single GPU without memory size constraints. `[10K+ downloads]`
https://github.com/grgalex/nvshare

PyXray Cross-language call graph construction for Python. Used for vulnerability reachability analysis and bloat detection.
https://github.com/grgalex/pyxray

Gasket Identifies foreign bridges in V8 embedders (Node.js, Deno), enabling security analysis across the language boundary.
https://github.com/gasket-tools/gasket

Skills

Programming:	C [systems, concurrency, function interposition, IPC], Python, Golang, Bash, Assembly [x86, AVR]
Program analysis:	Call graph construction, cross-language analysis, dynamic slicing, debloating
Binary analysis:	ELF internals, reverse engineering, Ghidra, binary provenance & version identification
Supply chain security:	Reachability analysis, cross-language SBOM generation
Runtime internals:	CPython C API & extension modules, V8 JavaScript engine
GPU & parallel computing:	CUDA, OpenMP, MPI, GPU virtualization/sharing, NVIDIA stack
Cloud & containers:	Kubernetes, Kubeflow, GCP/AWS, Docker, containerd, OCI [runc]
Miscellaneous:	Linux system administration, Git, TCP/IP, DNS, tcpdump
Languages:	Greek [native speaker], English [Cambridge C2 Proficiency], German [Goethe-Zertifikat B2], Chinese [HSK 3]